- **Park Brake Switch Input** - Each time the cluster detects ground on the red brake warning indicator driver circuit (park brake switch closed = park brake applied or not fully released) the brake indicator is illuminated solid. If a vehicle speed message is received by the cluster from the PCM over the PCI data bus indicating the vehicle is moving while the red brake warning indicator driver input is grounded, the brake indicator is flashed on and off repeatedly. Whether illuminated solid or flashing, the indicator remains illuminated until the red brake warning indicator driver input to the cluster is an open circuit (park brake switch open = park brake fully released), or until the ignition switch is turned to the Off position, whichever occurs first.
- **Brake Hydraulic System Malfunction** - Each time the cluster detects ground on the red brake warning indicator driver circuit (brake warning indicator switch closed = pressures in the two halves of the split brake hydraulic system are unequal) the brake indicator is illuminated solid. The indicator remains illuminated until the red brake warning indicator driver input to the cluster is an open circuit (brake warning indicator switch open = brake hydraulic system pressures are equal), or until the ignition switch is turned to the Off position, whichever occurs first.
- **Actuator Test** - Each time the instrument cluster is put through the actuator test, the brake indicator will be turned on, then off again during the bulb check portion of the test to confirm the functionality of the LED and the cluster control circuitry.

The park brake switch and the brake warning pressure switch are each connected in parallel between ground and the red brake warning indicator driver input to the instrument cluster so that each of their inputs will illuminate the indicator independently of the other. The park brake switch and brake warning indicator switch inputs to the instrument cluster can be diagnosed using conventional diagnostic tools and methods. (Refer to 5 - BRAKES/PARKING BRAKE - OPERATION). (Refer to 5 - BRAKES/ELECTRICAL/BRAKE PRESSURE SWITCH - OPERATION).

For further diagnosis of the brake indicator or the instrument cluster circuitry that controls the LED, (Refer to 8 - ELECTRICAL/INSTRUMENT CLUSTER - DIAGNOSIS AND TESTING). For proper diagnosis of the PCM, the PCI data bus, or the message inputs to the instrument cluster that control the brake indicator, a DRBIII® scan tool is required. Refer to the appropriate diagnostic information.

DIAGNOSIS AND TESTING - BRAKE INDICATOR

The diagnosis found here addresses an inoperative brake indicator condition. If there are problems with several indicators in the instrument cluster, (Refer to 8 - ELECTRICAL/INSTRUMENT CLUSTER - DIAGNOSIS AND TESTING). If the brake indicator stays on with the ignition switch in the On position and the park brake released, or comes on while driving, (Refer to 5 - BRAKES - DIAGNOSIS AND TESTING). If no brake system problem is found, the following procedures will help to locate a shorted or open circuit, or a faulty switch input. Refer to the appropriate wiring information. The wiring information includes wiring diagrams, proper wire and connector repair procedures, details of wire harness routing and retention, connector pin-out information and location views for the various wire harness connectors, splices and grounds.

WARNING: TO AVOID PERSONAL INJURY OR DEATH, ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, AIRBAG, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT.

INDICATOR ILLUMINATES DURING BULB TEST, BUT DOES NOT WHEN PARK BRAKE APPLIED

(1) Disconnect and isolate the battery negative cable. Disconnect the floor wire harness connector for the park brake switch from the switch terminal. Apply the parking brake. Check for continuity between the park brake switch terminal and a good ground. There should be continuity. If OK, go to Step 2. If not OK, replace the faulty park brake switch.

(2) Disconnect the headlamp and dash wire harness connector for the brake warning indicator switch from the switch terminals. Check for continuity between the red brake warning indicator driver (G9) circuit cavities of the floor wire harness connector for the park brake switch and the headlamp and dash wire harness connector for the brake warning indicator switch. There should be continuity. If not OK, repair the open red brake warning indicator driver (G9) circuit between the park brake switch and the brake warning indicator switch as required.

INDICATOR REMAINS ILLUMINATED - BRAKE SYSTEM CHECKS OK

(1) Disconnect and isolate the battery negative cable. Remove the instrument cluster from the instrument panel. Disconnect the instrument panel